# MICROSOFT AZURE

Cloud Computing Project

# Azure Logic Apps — No-Code Email Automation

*Automate Email Archiving to OneDrive Without Writing Any Code*

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|---|---|
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| **Course** | Cloud Computing |
| **Platform** | Microsoft Azure |
| **Service** | Azure Logic Apps |
| **Coding Required** | No Code |

## 1. Project Overview

| Field | Details |
|---|---|
| Project Name | Email to OneDrive Automation |
| Platform | Microsoft Azure |
| Service Used | Azure Logic Apps |
| Trigger | New Email in Outlook Inbox |
| Action | Save File in OneDrive |
| Hosting Plan | Consumption (Multi-tenant) |
| Coding Required | Zero — No Code Needed |
| Cost | Azure Free Credits |

## 2. Technologies Used

| Technology | Purpose |
|---|---|
| Microsoft Azure | Cloud Platform hosting all services |
| Azure Logic Apps | No-code workflow automation engine |
| Office 365 Outlook | Email source — acts as the trigger |
| Microsoft OneDrive | File storage — destination for saved emails |
| Azure Resource Group | Logical container for all project resources |

## 3. Architecture & Workflow

The project follows a event-driven architecture:

| TRIGGER | ACTION |
|---|---|
| New Email Arrives in Outlook Inbox | Create File in OneDrive |
| **INPUT:**<br>• Email Subject<br>• Email Body<br>• Sender Information | **OUTPUT:**<br>• File Name = Email Subject + .txt<br>• File Content = Email Body<br>• Saved in /EmailBackups |

## 4. Prerequisites

- Microsoft Azure Account — Sign up free at azure.microsoft.com/free
- Microsoft Outlook Account — outlook.com (personal or work)
- OneDrive Account — Personal or OneDrive for Business
- Web Browser — Google Chrome or Microsoft Edge

# 5. Step-by-Step Implementation

## PHASE 1 — Open Azure Portal

**Step 1** Open your web browser and navigate to: https://portal.azure.com
**Step 2** Sign in with your Microsoft / Outlook account credentials.
**Step 3** You will land on the Azure Dashboard — the main control panel.

## PHASE 2 — Create Resource Group

*A Resource Group is a logical container that holds all your Azure resources together.*

**Step 4** In the top search bar, type: Resource Groups
**Step 5** Click '+ Create' button.
**Step 6** Fill in the following details:

| Field | Value |
|---|---|
| Subscription | Azure Free Subscription |
| Resource Group Name | LogicAppRG |
| Region | Central India |

**Step 7** Click 'Review + Create' then click 'Create'.

**Resource Group created successfully!**

## PHASE 3 — Create Logic App

**Step 8** In the top search bar, type: Logic Apps
**Step 9** From the results, click 'Logic apps' — the 3rd option with the green icon. Do NOT select 'Logic Apps Templates' or 'Logic Apps Custom Connector'.
**Step 10** Click '+ Create' button.

## PHASE 4 — Select Hosting Plan

You will see the hosting plan selection screen with multiple options:

| Plan | Type | Select? |
|---|---|---|
| Multi-tenant | Consumption — Pay per use | YES |
| Workflow Service Plan | Standard — Dedicated | No |
| App Service Environment V3 | Standard — Enterprise | No |
| Hybrid | Standard — Complex | No |
| Managed (Preview) | Automation Project | No |

**Step 11** Click the radio button on 'Multi-tenant' under the Consumption column.
**Step 12** Click the 'Select' button at the bottom of the page.

**Why Multi-tenant?** Pay-per-operation model — nearly FREE for demos. No server setup needed. Best for beginners. Uses Azure free credits.

## PHASE 5 — Fill Logic App Details

**Step 13** Fill in the Logic App creation form with the following values:

| Field | Value |
|---|---|
| Subscription | Azure Free Subscription |
| Resource Group | LogicAppRG (select existing) |
| Logic App Name | EmailToOneDrive-App |
| Region | Central India |
| Enable Log Analytics | No (uncheck) |
| Plan Type | Consumption |

**Step 14** Click 'Review + Create'.
**Step 15** Review the summary, then click 'Create'.
**Step 16** Wait approximately 1 minute for deployment to complete.
**Step 17** Click 'Go to Resource' once deployment finishes.

## PHASE 6 — Design the Workflow

**Step 18** On the Logic App Overview page, click 'Logic App Designer' from the left-side menu.
**Step 19** You will see a blank workflow canvas ready for configuration.

### PART A — Add the Trigger (When email arrives)

**Step 20** Click 'Add a trigger' on the canvas.
**Step 21** In the search box, type: Outlook
**Step 22** Select 'Office 365 Outlook' from the results.
**Step 23** Under Triggers, select: 'When a new email arrives (V3)'
**Step 24** Click 'Sign In' and login with your Outlook / Microsoft account.
**Step 25** Configure the trigger settings as follows:

| Setting | Value |
|---|---|
| Folder | Inbox |
| Importance | Any |
| Include Attachments | No |

*This means: Every time a new email arrives in your Inbox, the Logic App workflow will start automatically.*

### PART B — Add the Action (Save to OneDrive)

**Step 26** Click '+ New Step' below the trigger block.
**Step 27** In the search box, type: OneDrive
**Step 28** Select 'OneDrive' (personal) or 'OneDrive for Business' as applicable.
**Step 29** Under Actions, select: 'Create File'
**Step 30** Click 'Sign In' and login with your OneDrive account.
**Step 31** Configure the action fields as follows:

| Field | How to Set It |
|---|---|
| Folder Path | Type manually: /EmailBackups |
| File Name | Click field > Click 'Dynamic Content' > Select 'Subject' > type .txt |
| File Content | Click field > Click 'Dynamic Content' > Select 'Body' |

**Dynamic Content:** Automatically pulls real-time data from the incoming email (Subject, Body, Sender) and inserts it into the action fields.

**Step 32** Click 'Save' at the top of the designer.

**Workflow Saved Successfully!**

## PHASE 7 — Test & Verify the Automation

**Step 33** Open Outlook and send a test email to yourself:

| Field | Value |
|---|---|
| To | your-email@outlook.com |
| Subject | LogicAppTest |
| Body | Hello! Azure Logic Apps is working successfully! |

**Step 34** Go back to Azure Portal > Logic App > Click 'Runs History' from the left menu.
**Step 35** Wait 1-2 minutes. You will see a run entry appear with:

| Field | Expected Value |
|---|---|
| Run Status | Succeeded |
| Trigger | When a new email arrives |
| Duration | ~2-3 seconds |
| Start Time | Just now |

**Step 36** Open your OneDrive > Navigate to /EmailBackups folder.
**Step 37** You will find a new file: LogicAppTest.txt containing your email body.

**Automation Working Successfully! Project Complete.**

## 7. Verification Checklist

| Item | Expected Result | Status |
|---|---|---|
| Logic App Status | Enabled and Running | Check in Overview |
| Trigger Configured | Outlook Inbox selected | Check in Designer |
| Action Configured | OneDrive Create File | Check in Designer |
| Dynamic Content | Subject and Body mapped | Check field values |
| Test Email Sent | Sent to own email | Do this manually |
| Run History | Shows Succeeded status | Check Runs History |
| File in OneDrive | File created in /EmailBackups | Open OneDrive |
| File Content | Matches email body text | Open the .txt file |

## 8. Real-World Use Cases

| Industry | Use Case |
|---|---|
| Corporate / Business | Auto-save important client emails to cloud storage |
| Healthcare | Archive patient inquiry and appointment emails automatically |
| Education / College | Save student assignment submission emails to OneDrive |
| E-Commerce | Log customer order confirmation emails automatically |
| HR Department | Archive job application and interview schedule emails |
| Finance / Accounting | Save invoice and payment emails to organized folders |
| Legal | Automatically backup all case-related email communications |

## 9. Project Summary

| Field | Details |
|---|---|
| Project Name | Email to OneDrive Automation |
| Cloud Platform | Microsoft Azure |
| Service Used | Azure Logic Apps |
| Hosting Plan | Consumption — Multi-tenant |
| Trigger | New Email in Outlook Inbox |
| Action | Create File in OneDrive /EmailBackups |
| Dynamic Content Used | Email Subject (file name) + Body (file content) |
| Total Steps | 43 Steps |
| Total Time | ~11 Minutes |
| Lines of Code Written | Zero |
| Cost | Nearly FREE (Azure Free Credits) |
| Difficulty Level | Beginner |

## 12. Conclusion

*This project demonstrates the power of cloud-based no-code automation using Azure Logic Apps. By connecting Microsoft Outlook and OneDrive through a simple trigger-action workflow, we automated a real-world business task — email archiving — in under 15 minutes without writing a single line of code. This showcases how Azure Logic Apps can save time, reduce manual effort, and streamline business processes across any industry. The pay-per-use pricing model makes it accessible and cost-effective for students, startups, and enterprises alike.*